Analysis of mean field games via Fokker-Planck-Kolmogorov equations: existence of equilibria

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Mean field games

Let us consider the problem (MFG):

$$\begin{cases} \mathbb{E}\left(\int_0^T f(t, X_t, \mu_t, \alpha(t, X_t, \mu_t))dt + g(X_T, \mu_T)\right) \longrightarrow \min_\alpha, \\ dX_t = b(t, X_t, \mu_t, \alpha(t, X_t, \mu_t))dt + \sigma(t, X_t, \mu_t, \alpha(t, X_t, \mu_t))dW_t, \\ \mu_t = \mathrm{Law}(X_t),\ \mu_0 = \nu. \end{cases}$$

"$\min_\alpha$" means that α is a minimum **under fixed μ** in a, σ, f, g,
that is, we solve the problem in two steps:

1) Find the set of minimum $\Phi(\mu)$ under fixed μ,

2) Solve the fixed point problem $\mu = \mathrm{Law}(X^\alpha)$ for some $\alpha \in \Phi(\mu)$.

Model of Flocking

As an example we can consider the following problem

$$\begin{cases} \dfrac{1}{2}\displaystyle\int_0^T \left(\alpha_t^2 + \left|X_t - \int y\,\mu_t(dy)\right|^2\right) dt \longrightarrow \min\limits_{\alpha}, \\ dX_t = \alpha_t dt + \sigma dW_t, \\ \mu_t = \mathrm{Law}(X_t),\ \mu_0 = \nu. \end{cases}$$

Hamilton—Jacobi–Bellman equation:

$$\begin{cases} \partial_t V + \Delta V - \dfrac{|\nabla V|^2}{2} + \dfrac{1}{2}\left|x - \displaystyle\int y\,\mu_t(dy)\right|^2 = 0, \quad V(x,T) = 0, \\ \partial_t \mu_t = \Delta \mu_t + \mathrm{div}(\nabla V\,\mu_t), \quad \mu_0 = \nu. \end{cases}$$

Stochastic maximum principle:

$$\begin{cases} dX_t = P_t dt + \sigma dW_t, \quad X_0 = x, \\ dP_t = \left(X_t - \displaystyle\int y\,\mu_t(dy)\right) dt + Z_t dW_t, \quad P_T = 0. \end{cases}$$

Superposition principle

Let $L^*_{A,b,\mu}\varphi = \partial_{x_i x_j}\left(a^{ij}(t,x,\mu)\varphi\right) - \partial_{x_i}\left(b^i(t,x,\mu)\varphi\right).$

Theorem (V. I. Bogachev, M. Röckner, S. V. Shaposhnikov, 2021): Let $\partial_t \mu = L^*_{A,b,\mu}\mu$. Suppose that

$$\int_0^T \int_{\mathbb{R}^d} \frac{\|A(t,x,\mu)\| + |\langle b(t,x,\mu), x\rangle|}{1+|x|^2}\,\mu_t(dx)dt < \infty.$$

Then there exists a probability space $(\Omega, \mathcal{F}, P, W_t)$ and an adapted process X such that

1. $dX_t = b(t, X_t, \mu)dt + (2A(t, X_t, \mu))^{1/2} dW_t,$
2. $\mu_t = P \circ X_t^{-1}$ for all $t \in [0,T]$.

The result is called **Ambrosio–Figalli–Trevisan Superposition Principle**.

Statement of the problem

The pair $(\mu_t dt, u)$ is called **a solution to** (MFG) if

1) $t \mapsto \mu_t$ is a continuous curve in the space of probability measures on $\mathbb{R}^d$ with the weak topology,

2) the mapping $u\colon \mathbb{R}^d \times [0,T] \to U \subset \mathbb{R}^{d_1}$ is Borel,

3) $\mu = \mu_t\, dt$ is a solution to the Cauchy problem

$$\begin{cases} \partial_t \mu_t = \partial_{x_i x_j}\left(a^{ij}\mu_t\right) - \partial_{x_i}\Big(\left(b^i + (Qu)^i\right)\mu_t\Big), \\ \mu_0 = \nu, \end{cases}$$

where $a^{ij} = a^{ij}(x,t,\mu)$, $b^i = b^i(x,t,\mu)$ and $Q = Q(x,t,\mu)$.

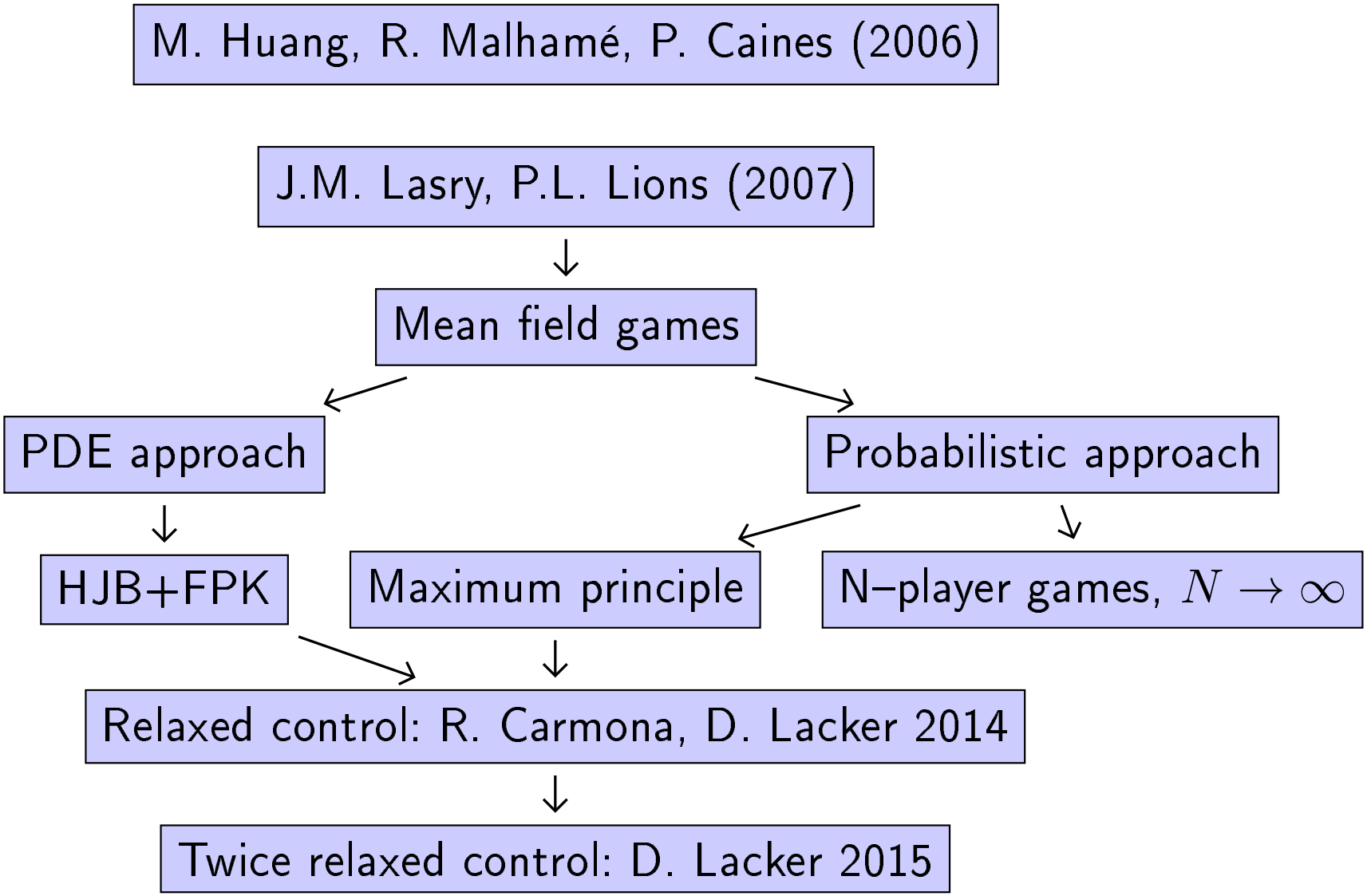
Statement of the problem

4) the following inequality is fulfilled

$$\int_0^T \int_{\mathbb{R}^d} f(u(x,t),x,t,\mu)\mu_t(dx)\,dt + \int_{\mathbb{R}^d} g(x,\mu)\mu_T(dx) \leq$$
$$\leq \int_0^T \int_{\mathbb{R}^d} f(v(x,t),x,t,\mu)\sigma_t(dx)\,dt + \int_{\mathbb{R}^d} g(x,\mu)\sigma_T(dx).$$

whenever the pair (σ, v) satisfies the conditions $1), 2)$ and $3)$ **with the measure μ in the coefficients**.

History of existence theorems



Major improvements

We consider the following case

1. Coefficients can depend on the entire measure $\mu_t dt$,

2. $b(t, x, \mu), Q(t, x, \mu)$ can be non-Lipschitz in x,

3. $Q(t, x, \mu)$ can depend on x even when the control set U is unbounded.

Lyapunov function

Let $V \in C^2(\mathbb{R}^d)$, $V \geq 0$ and $\lim_{|x|\to+\infty} V(x) = +\infty$.

Example: $V(x) = (1 + |x|^2)^{m/2}$, where $m > 0$.

Let us denote by $\mathcal{M}_R(V)$ the set of continuous curves $t \mapsto \mu_t$ in the space of probability measures on $\mathbb{R}^d$ with the weak topology such that

$$\sup_{t\in[0,T]} \int_{\mathbb{R}^d} V(x)\, \mu_t(dx) \leq R.$$

Let us also introduce the set $\mathcal{M}(V) = \cup_{R>0}\mathcal{M}_R(V)$.

V–weak convergence

Let us say that the sequence $\mu^n = \mu_t^n \, dt \in \mathcal{M}(V)$ **converges** V **– weakly** to the measure $\mu = \mu_t \, dt \in \mathcal{M}(V)$ if

$$\lim_{n \to \infty} \int_{\mathbb{R}^d} \zeta(x) \mu_t^n(dx) = \int_{\mathbb{R}^d} \zeta(x) \mu_t(dx)$$

for all $t \in [0, T]$ and every continuous function ζ such that

$$\lim_{|x| \to +\infty} \frac{\zeta(x)}{V(x)} = 0.$$

H1: Local conditions

(**H1.1**) For every open ball $B \subset \mathbb{R}^d$ and every $R > 0$

$$\sup_{x \in B, t \in [0,T], \mu \in \mathcal{M}_R(V)} \Big(|a^{ij}(x,t,\mu)| + |b^i(x,t,\mu)| + |q^{ml}(x,t,\mu)| \Big) < \infty.$$

(**H1.2**) For every measure $\mu \in \mathcal{M}(V)$ and all $t \in [0,T]$ the functions

$$x \to a^{ij}(x,t,\mu), \quad x \to b^i(x,t,\mu), \quad x \to q^{ml}(x,t,\mu)$$

are continuous on $\mathbb{R}^d$.

(**H1.3**) For every open ball B, all $t \in [0,T]$ and every number $R > 0$ the V–weak convergence of $\mu^n \in \mathcal{M}_R(V)$ to $\mu \in \mathcal{M}_R(V)$ implies the equality

$$\lim_{n \to \infty} \sup_{x \in B} \Big(|a^{ij}(x,t,\mu^n) - a^{ij}(x,t,\mu)| + |b^i(x,t,\mu^n) - b^i(x,t,\mu)| + |q^{ml}(x,t,\mu^n) - q^{ml}(x,t,\mu)| \Big) = 0.$$

Functional growth control functions

Growth in x and μ: Let W is a continuous function on $\mathbb{R}^d$ such that

$$0 \leq W(x) \leq V(x), \qquad \lim_{|x| \to +\infty} \frac{W(x)}{V(x)} = 0.$$

Growth in u: Let h is a convex increasing function on $[0, +\infty)$ such that

$$h(0) = 0, \qquad \lim_{v \to +\infty} \frac{h(v)}{v} = +\infty.$$

Let us denote by h^* the Legendre transform of h, i.e.

$$h^*(v) = \sup_{p \geq 0} \big(pv - h(p) \big).$$

H2: Global conditions

(**H2.1**) There exists a number $C_L > 0$ such that the estimate

$$L_\mu V(x,t) + h^*\big(|Q^\top(x,t,\mu)\nabla V(x)|\big) \leq$$
$$C_L V(x) + C_L \int_{\mathbb{R}^d} V(y)\mu_t(dy) + C_L \sup_{t\in[0,T]} \int_{\mathbb{R}^d} W(y)\mu_t(dy)$$

holds for every $\mu \in \mathcal{M}(V)$, $x \in \mathbb{R}^d$ and all $t \in [0,T]$.

H2: Global conditions

(**H2.2**) For every measure $\mu \in \mathcal{M}(V)$ there exist a number $C_1(\mu) > 0$ and a nonnegative function Θ such that for all $x, y \in \mathbb{R}^d$ and $t \in [0, T]$ we have

$$\operatorname{tr}\left(\left(\sqrt{A(x,t,\mu)} - \sqrt{A(y,t,\mu)}\right)^2\right) + \left\langle b(x,t,\mu) - b(y,t,\mu), x - y\right\rangle \leq$$
$$\leq C_1(\mu)\Big(1 + V(x) + V(y)\Big)|x-y|^2,$$

and

$$\|Q(x,t,\mu) - Q(y,t,\mu)\| \leq \Big(\Theta(x,t) + \Theta(y,t)\Big)|x-y|.$$

(**H2.3**) For every measure $\mu \in \mathcal{M}(V)$ there exists a number $C_2(\mu) > 0$ such that for all $x \in \mathbb{R}^d$ and $t \in [0, T]$ we have

$$\|A(x,t,\mu)\| + |b(x,t,\mu)| + h^*(\|Q(x,t,\mu)\|) + h^*(\Theta(x,t)) \leq C_2(\mu)V(x).$$

H3: Conditions on f and g

(**H3.1**) The function g is continuous in x and there exists a number $C_g > 0$ such that for all $x \in \mathbb{R}^d$ and every $\mu \in \mathcal{M}(V)$ we have

$$|g(x,\mu)| \leq C_g\Big(W(x) + \sup_{t\in[0,T]} \int_{\mathbb{R}^d} W(y)\mu_t(dy)\Big).$$

(**H3.2**) There exist numbers $C_h > 1$ and $C_f > 0$ such that the inequalities

$$h(|u|) - C_f\Big(W(x) + \sup_{t\in[0,T]} \int_{\mathbb{R}^d} W(y)\mu_t(dy)\Big) \leq f(u,x,t,\mu) \leq$$
$$\leq C_h h(|u|) + C_f\Big(W(x) + \sup_{t\in[0,T]} \int_{\mathbb{R}^d} W(y)\mu_t(dy)\Big)$$

hold for every $\mu \in \mathcal{M}(V)$, $u \in \mathbb{R}^{d_1}$, $x \in \mathbb{R}^d$ and $t \in [0,T]$.

H3: Conditions for f and g

(**H3.3**) For all open balls $B \subset \mathbb{R}^d$, $B_1 \subset \mathbb{R}^{d_1}$ and every number $R > 0$ the V–weak convergence of $\mu^n \in \mathcal{M}_R(V)$ to $\mu \in \mathcal{M}_R(V)$ implies the equality

$$\lim_{n\to\infty} \sup_{x\in B} |g(x, \mu^n) - g(x, \mu)| = 0$$

and for every $t \in [0, T]$ the equality

$$\lim_{n\to\infty} \sup_{u\in B_1\cap U, x\in B} |f(u, x, t, \mu^n) - f(u, x, t, \mu)| = 0.$$

(**H3.4**) The function f is continuous in (x, u) and convex in u.

Main result

Theorem (S.V. Shaposhnikov, S., 2025): Assume that the conditions $(\mathrm{H1}), (\mathrm{H2}), (\mathrm{H3})$ are fulfilled and $V \in L^1(\nu)$. Then there exists a measure $\mu = \mu_t\, dt \in \mathcal{M}(V)$ and a Borel function $u(x,t)$ such that

(i) $h(|u(x,t)|)$ is integrable with respect to μ,

(ii) $\mu = \mu_t\, dt$ is a solution to the Cauchy problem

$$\partial_t \mu_t = L^*_{\mu,u(x,t)}\mu_t, \quad \mu_0 = \nu,$$

(iii) the following inequality is fulfilled

$$\int_0^T \int_{\mathbb{R}^d} f(u(x,t),x,t,\mu)\,\mu_t(dx)\,dt + \int_{\mathbb{R}^d} g(x,\mu)\mu_T(dx) \leq$$
$$\leq \int_0^T \int_{\mathbb{R}^d} f(v(x,t),x,t,\mu)\,\sigma_t(dx)\,dt + \int_{\mathbb{R}^d} g(x,\mu)\sigma_T(dx)$$

whenever $h(|v(x,t)|)$ is integrable with respect to $\sigma_t\, dt$.

Example 1: D. Lacker conditions

Let $V(x) = 1 + |x|^2$, $W(x) = |x|$ and $h(v) = Cv^2$, where $C > 0$.

Suppose that for all $x, y \in \mathbb{R}^d$, $t \in [0, T]$ and every $\mu \in \mathcal{M}(V)$

1. $\|\sqrt{A(x,t,\mu)} - \sqrt{A(y,t,\mu)}\| + \|Q(x,t,\mu) - Q(y,t,\mu)\| + |b(x,t,\mu) - b(y,t,\mu)| \leq C_1|x-y|$,

2. $\|\sqrt{A(0,t,\mu)}\| + |b(0,t,\mu)| \leq C_2 + C_2 \int_{\mathbb{R}^d} |y|\mu_t(dy)$, $\|Q\| \leq C_2$,

3. $|g(x,\mu)| \leq C_3\big(1 + |x|\big) + C_3 \int_{\mathbb{R}^d} |y|\mu_t(dy)$,

4. $C|u|^2 - C_4\big(1 + |x|\big) - C_4 \int_{\mathbb{R}^d} |y|\mu_t(dy) \leq f(u,x,t,\mu) \leq$

$$\leq C_5|u|^2 + C_4\big(1 + |x|\big) + C_4 \int_{\mathbb{R}^d} |y|\mu_t(dy).$$

If, in addition, the continuity with respect to measures holds, then **the conditions of the theorem are fulfilled**.

Example 2: Non-Lipschitz drift

Let $V(x) = (1+|x|^2)^{m/2}$, $W(x) = |x|^p$ and $h(v) = Cv^2$, where $m \geq 2$, $1 \leq p < m$ and $C > 0$.

Suppose that for all $x, y \in \mathbb{R}^d$, $t \in [0,T]$ and every $\mu \in \mathcal{M}(V)$

1. $\|\sqrt{A(x,t,\mu)} - \sqrt{A(y,t,\mu)}\| + \|Q(x,t,\mu) - Q(y,t,\mu)\| \leq C_1|x-y|$

$$\|\sqrt{A(x,t,\mu)}\| \leq C_1\big(1+|x|\big), \quad \|Q(x,t,\mu)\| \leq C_1\big(1+|x|^{1-\varepsilon}\big),$$

2. $|b(x,t,\mu) - b(y,t,\mu)| \leq C_2(1+|x|^{m-1}+|y|^{m-1})|x-y|$,

$\langle b(x,t,\mu), x\rangle \leq C_2 - C_3|x|^{m+1}$,

3. $|g(x,\mu)| \leq C_4\big(1+|x|^p\big) + C_4 \int_{\mathbb{R}^d} |y|^p \mu_t(dy)$,

4. $C|u|^2 - C_5\big(1+|x|^p\big) - C_5 \int_{\mathbb{R}^d} |y|^p \mu_t(dy) \leq f(u,x,t,\mu) \leq$

$$\leq C_6|u|^2 + C_5\big(1+|x|^p\big) + C_5 \int_{\mathbb{R}^d} |y|^p \mu_t(dy).$$

If, in addition, the continuity with respect to measures holds, then **the conditions of the theorem are fulfilled**.

The talk is based on

S.V. Shaposhnikov, D.V. Shatilovich, Analysis of mean field games via Fokker–Planck–Kolmogorov equations: existence of equilibria, Nonlinear Analysis 270 (2026)

Thank you!